

Program : Diploma in Instrumentation Engineering	
Course Code : 6089	Course Title: Virtual Instrumentation Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week : 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience for the students with the programming concepts of LabVIEW.
- To equip Students to develop logics, create programs, applications in VI / Matlab software

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I & II	1 & 2
Basic Knowledge of IT skills and basic problem solving		Introduction to IT systems	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop simple Lab view programs	7	Applying
CO2	Develop VI programs using for loop, while loop, shift register, formula node and measure real world signals using NI myDAQ	16	Applying
CO3	Develop MATLAB programs related to Industrial Electronics	11	Applying
CO4	Develop MATLAB Programs related to Process Control Applications	8	Applying
	Lab Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement simple LabVIEW program		
M1.01	Implement arithmetic operations using numerical control	1	Applying
M1.02	Implement the expression $3x^2 + 2x + 5$ and display the output using a numeric indicator. Input x from numeric control	1	Applying
M1.03	Simulation of sinusoidal signal and display in the wave form graph. Change amplitude and frequency	1	Applying
M1.04	Implement Temperature conversions (Degree Celsius, Degree Kelvin, Fahrenheit)	1	Applying
M1.05	Implement a program to compare temperature with a set point and if it exceeds the limit, an indicator is ON.	1	Applying
M1.06	Develop a VI to find the average of two numbers and convert a section of a VI into a subVI Create a subVI to convert radians to degrees.	2	Applying
CO2	Develop VI programs using for loop, while loop, shift register, formula node and measure real world signals using NI myDAQ		
M2.01	Develop VI programs using for loop and while loop	2	Applying
M2.02	Develop VI programs using shift register and feedback node	3	Applying

M2.03	Develop a VI program to perform Array function	3	Applying
M2.04	Develop a VI program to illustrate cluster and structure	4	Applying
M2.05	Develop a multimeter using NI myDAQ	2	Applying
M2.06	Develop a program to measure current and voltage using NI myDAQ	2	Applying
	Lab test-I	2	
CO3	Develop MATLAB Programs related to Industrial Electronics		
M3.01	Simulate uncontrolled and controlled rectifier	5	Applying
M3.02	Simulate and study the operation of Inverter	2	Applying
M3.03	Simulate and study the operation of Converter	2	Applying
M3.04	Simulate and study the operation of Chopper	2	Applying
CO4	Develop MATLAB Programs related to Process Control Applications		
M4.01	Develop Low pass & High pass filter	2	Applying
M4.02	Simulate P,PI,PD,PID control using MATLAB	2	Applying
M4.03	Simulate uncontrolled and controlled rectifier	2	Applying
M4.04	Develop Step, Ramp, Impulse responses of First order system	2	Applying
	Lab test-II	1	

Text / References:

T/R	Book Title/Author
R1	Jovitha Jerome, Virtual Instrumentation using LabVIEW, PHI
R2	Richard Jennings, Lab VIEW Graphical Programming, McGraw-Hill Fifth Edition
R3	Sanjay Gupta, Virtual Instrumentation using LABVIEW, McGraw Hill second edition

Online Resources:

Sl. No	Website Link
1	https://www.academia.edu/9455052/Virtual_Instruments_using_LabView_by_-_Jovitha_Jerome
2	https://www.vidyarthiplus.com/vp/attachment.php?aid=24251
3	https://wwwusers.ts.infn.it/~rui/univ/Acquisizione_Dati/Lezioni/VIII%20-%20Labview%20-%20Introduction/LabVIEW%20Introduction-SixHour.pdf
4	https://manual.eg.poly.edu/index.php/Introduction_to_LabVIEW
5	http://www.dsod.p.lodz.pl/materials/EM07_IFE.pdf
6	https://www.ni.com/pdf/manuals/320999e.pdf
7	http://fy.chalmers.se/~f7xlh/LV_Kurs/Measur_Man_2000.pdf